

# SCHOLAR: A Local-First Research Assistant for Scientific Papers

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## Abstract

Scientific-document assistants can answer questions over research papers, but verifying their answers still requires readers to locate and inspect the supporting evidence. We present SCHOLAR, a local-first scientific-document assistant designed around inspectable citation grounding. SCHOLAR preserves source identity throughout document processing, allowing citations in generated answers to resolve directly to the corresponding paragraph, table, or figure region in the original PDF. The system combines lexical and dense retrieval with local answer generation, post-generation citation checks, and an auditable evidence trace. On a retrospective benchmark of 200 questions across 10 scientific papers, fused retrieval achieves a gold-page Hit@1 of 0.85 and an MRR of 0.905. Across 45 questions sampled from 9 papers, five human raters assigned a mean grounding score of 82.6%. A separate automated audit provides additional diagnostics of grounding and citation quality. SCHOLAR turns generated citations into direct, inspectable links to source evidence while keeping the question-answering workflow local. The demo and source code are available at <https://github.com/prithvi-kaizen/Scholar>.

## 1 Introduction

Scientific papers are not flat text documents. Their claims are often distributed across prose, tables, figures, captions, and other layout-dependent elements, with the surrounding page providing context that may be needed to interpret a result correctly. Question-answering systems can reduce the effort required to search through this material, but they introduce a second task for the reader: checking the generated answer against the paper itself. A citation may identify a source or retrieved passage, yet the reader may still need to locate that evidence in the original PDF and inspect it in context.

Retrieval-augmented generation (RAG) addresses part of this problem by supplying models

with external evidence during generation (Lewis et al., 2020), while work on citation-grounded generation studies how generated statements can be attributed to supporting sources (Gao et al., 2023). These approaches establish a connection between an answer and retrieved evidence, but that connection does not necessarily preserve the evidence’s location in the source document. Scientific-document pipelines commonly convert PDFs into retrieval units, and information such as page identity and spatial position can be separated from the text used for generation. This matters when the relevant evidence is a particular table entry, figure, caption, or paragraph rather than an interchangeable block of text.

For source verification, provenance therefore needs to survive the full path from document ingestion to the user interface. SCHOLAR is built around this requirement. Each evidence object retains its document, page, and region identity as it moves through retrieval and answer generation. Citations in the generated response refer to these evidence objects, allowing the application to resolve them back to their original locations in the PDF rather than asking the language model to reconstruct that information. The reader can then move from a claim in the answer to the corresponding source region and inspect the surrounding context directly. The document-processing and answering pipeline runs locally, so the same workflow can also be used with material that users may not want to send to an external service, such as unpublished manuscripts or review drafts.

Previous work has studied the individual problems that meet in this setting. Scientific question-answering and verification benchmarks examine reasoning over research papers and their supporting evidence (Dasigi et al., 2021; Wadden et al., 2020), while multimodal benchmarks extend scientific QA to figures, tables, and multiple documents (Pramanick et al., 2024; Li et al., 2024). Other work inves-

investigates whether generated responses are adequately grounded in retrieved evidence (Asai et al., 2024). SCHOLAR focuses on a complementary systems question: how to preserve an inspectable path from a generated claim to the location of its evidence in the original scientific document. We discuss the relationship to these systems in Section 2.

This demonstration makes three contributions:

1. **Source-preserving document representation:** evidence retains document, page, and region identity throughout the answering pipeline, keeping citation provenance under application control.
2. **Inspectable scientific question answering:** generated citations resolve to the corresponding text, figure, or table region in the original PDF, with retrieval and verification actions available through an audit trace.
3. **Local and reproducible evaluation:** an open implementation evaluated for evidence retrieval, grounding, citation quality, and end-to-end execution on consumer hardware.

## 2 Related Work

**Scientific QA and reading systems.** QASPER links answers to evidence in full research papers, SPIQA extends scientific QA to figures and tables, M3SciQA adds multimodal multi-document reasoning, and SciFact studies scientific claim verification (Dasigi et al., 2021; Pramanick et al., 2024; Li et al., 2024; Wadden et al., 2020). Interactive systems span academic-search question answering (Zhao and Lee, 2020), literature synthesis with citations (Lála et al., 2023; Skarliniski et al., 2024), and augmented reading interfaces (Lo et al., 2024). SCHOLAR focuses specifically on preserving a citation’s mapping to a spatial region in the original PDF while keeping the answering workflow local.

**Attribution and inspectable evidence.** Attribution work evaluates whether generated content is supported by an identified source (Rashkin et al., 2023; Gao et al., 2023), while Self-RAG integrates retrieval with self-reflection (Asai et al., 2024). Recent systems expose evidence at sentence, paragraph, or token level, including SciTrue, ClinicalTrialsHub, and TrialsSummarizer (Tan et al., 2026; Park et al., 2026; Ramprasad et al., 2023). SCHOLAR complements these systems by retaining page and bounding-box coordinates so citations

can be inspected in their original layout. Its lexical and numerical checks are diagnostic rather than semantic-entailment judgments.

### Document representation and local operation.

GROBID, Nougat, and Docling convert scholarly PDFs into structured representations (Lopez, 2009; Blecher et al., 2024; Auer et al., 2024), while ColPali retrieves directly over page-image embeddings (Faysse et al., 2025). Recent EACL demonstrations apply document structure to complex tables and heterogeneous reports (Santacrose et al., 2026; Vaghefi et al., 2026); InkSight uses hosted generation through OpenRouter (Sakhovskiy et al., 2026). SCHOLAR instead combines BM25, dense retrieval, and reciprocal-rank fusion (Robertson and Zaragoza, 2009; Cormack et al., 2009) while keeping parsing, generation, verification, and source coordinates under local application control.

## 3 System Architecture

SCHOLAR is organized around one requirement: evidence retrieved from a scientific paper should remain traceable to its location in the original document throughout question answering. A local FastAPI backend handles ingestion, retrieval, generation, verification, and trace persistence, while a Next.js/React frontend renders answers beside the source PDF. Figure 1 summarizes the pipeline.

### 3.1 Source-preserving document representation

During ingestion, Docling (Auer et al., 2024) is used when its local resources are available, with PyMuPDF as a fallback. Parser-produced text blocks, tables, and visual elements are retained as retrieval units rather than passed through a second fixed-length chunker. Each evidence object stores its document identifier, page, modality, content, and a normalized bounding box. Figures and tables retain their visual regions, while page images are stored separately for visual retrieval and inspection.

Page and bounding-box information remain explicit application metadata rather than being reconstructed later. The same provenance-bearing evidence objects therefore pass through retrieval, generation, verification, and interface navigation.

### 3.2 Evidence retrieval

Lexical retrieval uses BM25 (Robertson and Zaragoza, 2009), while dense retrieval uses lo-

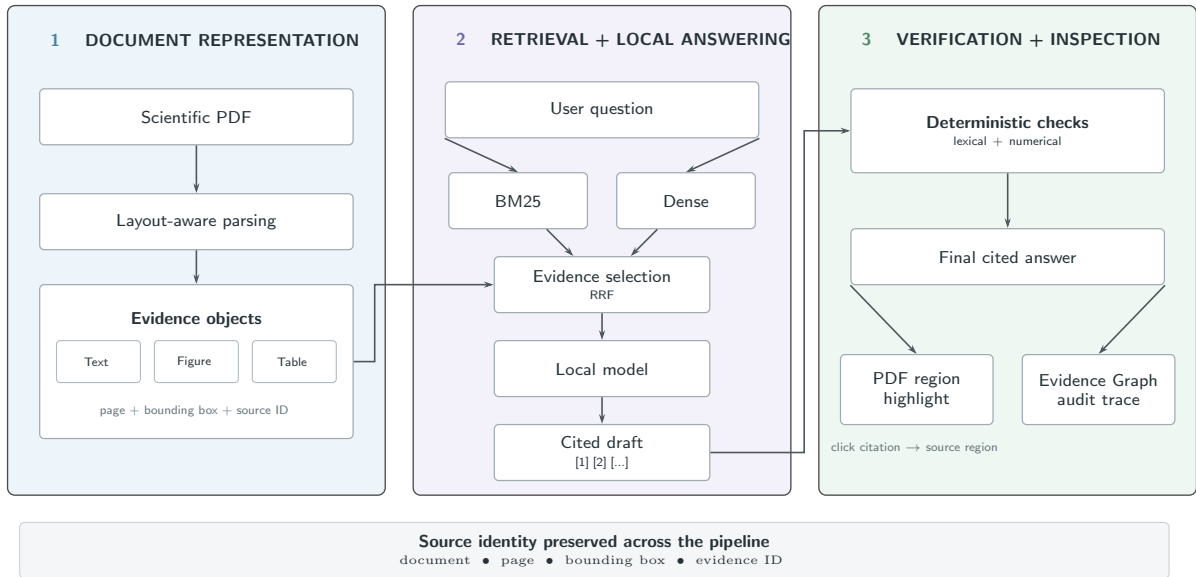


Figure 1: **SCHOLAR architecture.** A scientific PDF is converted into source-scoped evidence objects that retain page and bounding-box metadata. Lexical, dense, and optional visual retrieval select evidence for local answer generation. Model-visible evidence identifiers are bound deterministically to source objects, checked after generation, and resolved back to highlighted PDF regions.

cal all-MiniLM-L6-v2 embeddings (Reimers and Gurevych, 2019). Figure and table crops or full pages can also enter the candidate pool when visual evidence is relevant. Figure/table intent is detected with keyword, regular-expression, section, and modality heuristics.

Ranked outputs are combined using reciprocal-rank fusion (Cormack et al., 2009) with  $k = 60$  and passed through a local cross-encoder reranker when available, with a deterministic fallback otherwise. At most seven evidence items enter the generation prompt. The retrieval stack is conventional by design; its role is to select evidence without discarding the source metadata attached during ingestion.

### 3.3 Local answer generation and citation binding

Selected evidence is serialized with temporary identifiers such as [E1] and [E2], together with source label, page, chunk ID, and supporting text. Generation runs locally through Ollama (Ollama, 2026), using qwen3.5:9b (Qwen Team, 2026) by default; selected images can also be supplied when the configured model supports visual input.

Before generation, the application maps each temporary identifier to its source object. Returned identifiers are normalized into numbered citations and resolved through this map; unknown identi-

fiers are discarded. The model therefore generates the answer, while page identity, coordinates, and citation resolution remain under deterministic application control.

### 3.4 Verification and inspection

Post-generation checks compare claims with cited evidence using lexical overlap, polarity terms, and numerical consistency. Depending on repair mode, the system may remap a citation, narrow a partially supported claim, or remove unsupported material. These checks are diagnostic rather than semantic-entailment judgments.

Each request produces a local JSON trace containing retrieval metadata, evidence, generated and normalized answers, citation records, verifier actions, timings, hardware metadata, and final status; the assembled prompt is represented by a hash. The Evidence Graph exposes the evidence used during answering, while citation clicks use retained page and bounding-box metadata to navigate to and highlight the corresponding PDF region.

With required models and parser assets installed, parsing, embedding, retrieval, generation, verification, and trace storage can run locally. Strict-local mode restricts model endpoints to local services and disables external acquisition features at the application level; if local Docling assets are unavailable, ingestion falls back to PyMuPDF rather

than downloading them. This is an application-level offline mode rather than an operating-system network-isolation guarantee.

## 4 Interactive Demonstration

The demonstration focuses on the interaction that motivates SCHOLAR: moving from a generated claim to the evidence that supports it in the original scientific document. Rather than presenting citations only as textual references, the interface keeps the answer and source paper connected throughout the reading workflow. Figure 2 shows the main workspace.

### 4.1 Demo workflow

A demonstration begins by loading and indexing a scientific PDF. The user then asks a question in natural language and receives a response whose citations are linked to the evidence objects selected during retrieval. Selecting a citation moves the document view to the corresponding page and highlights the retained source region when bounding-box information is available.

The same interaction can be followed beyond the final citation. The Evidence Graph provides an ordered view of the evidence used during answering, including its page, modality, and role in the evidence path. The user can move from these entries back to the corresponding source pages and can export the resulting reasoning report in Markdown or  $\text{\LaTeX}$  format. The graph is intended as an inspection aid rather than a claim that the displayed ordering constitutes a verified reasoning chain.

### 4.2 Representative scenarios

The live demonstration covers three types of questions that exercise different parts of the system.

**Text and numerical evidence.** A user can ask a question such as, “What improvement does the proposed method report over baseline X?” SCHOLAR retrieves relevant textual and tabular evidence, generates a cited response, and allows the reader to open the cited source region directly. This scenario demonstrates how a numerical statement in the answer can be checked against the experiment description or table from which it was retrieved.

**Figures and tables.** The user can instead ask a question tied to visual material, such as, “What does Figure 3 show about the ablation?” The retrieval pipeline can include figure or table evidence

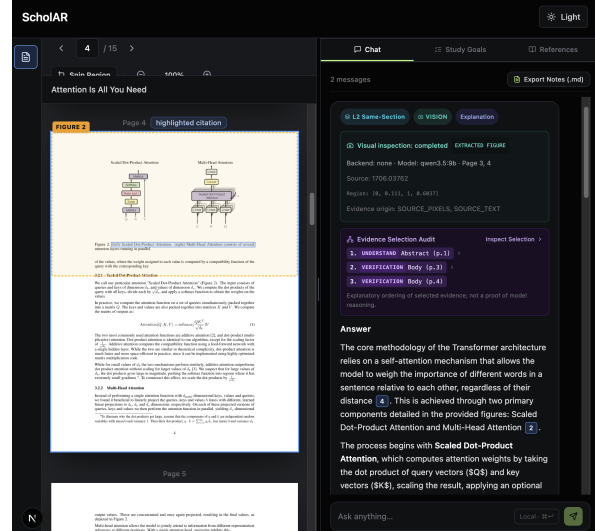


Figure 2: **SCHOLAR demonstration interface.** The generated answer and source document are shown in the same workspace. Selecting a citation moves the document view to the corresponding page and highlights the stored evidence region, allowing the reader to compare the generated claim with its source context directly.

together with associated textual context. When the resulting citation is selected, the interface returns the reader to the corresponding region of the paper so that the generated description can be compared with the original visual evidence.

**Evidence-path inspection.** The third scenario focuses on how an answer was assembled rather than on the answer alone. After receiving a cited response, the user opens the Evidence Graph to inspect the evidence objects associated with the run, including their pages, modalities, and roles. This provides a second route from the generated response back to the document and makes the evidence used during answering visible without treating the model output itself as proof of correctness.

### 4.3 Intended use

SCHOLAR is intended for situations in which reading and verification happen together. A researcher comparing experimental results can move from a reported number to its source table; a peer reviewer can inspect the evidence behind a generated summary; and a student reading a technical paper can follow a cited explanation back to the relevant paragraph or figure.

During the live demonstration, we show the complete path from document ingestion to evidence inspection: loading a paper, asking text and visual questions, selecting citations, inspecting

highlighted source regions, opening the Evidence Graph, and exporting the corresponding reasoning report. The screencast follows the same workflow so that the interaction can be reproduced outside the live session.

## 5 Evaluation

We evaluate SCHOLAR along four dimensions: page-level evidence retrieval, human assessment of generated answers and citations, a post-hoc automated audit, and query-time execution on consumer hardware. These evaluations target different parts of the system and should not be interpreted as a single combined benchmark.

### 5.1 Evidence retrieval

We evaluate page-level evidence retrieval on a benchmark of 200 questions drawn from 10 scientific papers. Each question is associated with a gold evidence page. We compare five retrieval configurations using Hit@1, Hit@5, mean reciprocal rank (MRR), and nDCG@5.

| Retriever    | H@1        | H@5        | MRR         | nDCG        |
|--------------|------------|------------|-------------|-------------|
| Keyword      | .71        | .91        | .796        | .757        |
| BM25         | .84        | .97        | .892        | <b>.855</b> |
| Dense        | .76        | .95        | .838        | .796        |
| BM25 + Dense | <b>.85</b> | <b>.98</b> | <b>.905</b> | .853        |
| + Modality   | .59        | .95        | .734        | .714        |

Table 1: **Page-level retrieval results.** Results are computed on 200 questions from 10 papers. BM25 + Dense uses reciprocal-rank fusion; the final row additionally applies the modality heuristic. Bold indicates the best value per metric.

BM25 retrieves the gold page at rank one for 84% of queries, with an MRR of 0.892. Combining BM25 and dense retrieval with reciprocal-rank fusion gives the highest Hit@1, Hit@5, and MRR, reaching 0.85, 0.98, and 0.905, respectively. The improvement over BM25 is small, indicating that lexical retrieval is already a strong baseline for this collection.

Adding the current modality heuristic reduces Hit@1 from 0.85 to 0.59. We therefore retain visual evidence as an available retrieval and inspection channel without treating the present modality heuristic as an additional ranking improvement.

### 5.2 Human evaluation

We conducted a human evaluation of answer quality and citation grounding on 45 questions drawn

from 9 scientific papers. Five evaluators independently rated the same 45 cases, yielding 225 case-level evaluations.

For each case, 5 evaluators inspected the question, generated answer, citations, and cited source evidence. They scored correctness, completeness, grounding, citation quality, and utility on five-point scales.

| Dimension        | Mean | $\geq 4$ | $\alpha$ |
|------------------|------|----------|----------|
| Correctness      | 4.16 | 79.6%    | -0.05    |
| Completeness     | 4.13 | 78.7%    | -0.01    |
| Grounding        | 4.13 | 79.6%    | -0.03    |
| Citation quality | 3.96 | 70.2%    | -0.02    |
| Utility          | 4.19 | 79.1%    | -0.07    |

Table 2: **Human evaluation on 45 questions from 9 papers.** Means are computed over 225 ratings on a five-point scale.  $\geq 4$  reports the proportion of ratings of four or five.  $\alpha$  is ordinal Krippendorff’s agreement coefficient.

Average ratings are above 3.9 in every dimension. Correctness, completeness, grounding, and utility average above 4 out of 5, while citation quality averages 3.96. Across the five dimensions, between 70.2% and 79.6% of individual ratings are at least four.

Inter-rater agreement remains low, with Krippendorff’s  $\alpha$  ranging from -0.07 to -0.01. The human results should therefore be interpreted as a distribution of evaluator judgments rather than strong consensus about individual answers.

At the citation level, evaluators produced 675 judgments. Of these, 472 (69.9%) are supported, 119 (17.6%) partially supported, 82 (12.1%) unsupported, and 2 (0.3%) unverifiable. Overall, 87.6% fall into the supported or partially supported categories.

### 5.3 Automated audit and repair

As a separate diagnostic, we analyze 150 completed cases using a post-hoc LLM-assisted audit. The retained evaluation artifacts contain categorical judgments for correctness, completeness, grounding, and citation quality. They do not provide a provider receipt that independently verifies the identity of the executing judge. We therefore treat these results as diagnostic labels rather than human-verified ground truth.

Correctness is evaluated on 113 reference-eligible cases. Of these, 59 are labeled correct, 33 partially correct, 20 incorrect, and one not judge-

able. Completeness, grounding, and citation quality are evaluated over all 150 cases.

| Dimension                | Pos. | Part. | Neg. | N/A |
|--------------------------|------|-------|------|-----|
| Correctness <sup>a</sup> | 52.2 | 29.2  | 17.7 | 0.9 |
| Completeness             | 55.3 | 16.0  | 27.3 | 1.3 |
| Grounding                | 74.7 | 17.3  | 6.7  | 1.3 |
| Citation quality         | 73.3 | 18.7  | 6.7  | 1.3 |

Values are percentages. <sup>a</sup>Correctness uses 113 reference-eligible cases; all other rows use 150 cases.

Table 3: **Post-hoc LLM-assisted audit.** The retained artifacts do not independently verify the executing judge’s identity; results are therefore treated as diagnostic labels rather than human ground truth.

Grounding is labeled positive in 112 of 150 cases (74.7%), and citation quality is positive in 110 cases (73.3%).

We additionally examine the deterministic repair stage on 30 paired answerable cases. Reference-unit overlap F1 increases from 0.159 before repair to 0.224 afterward. Page-level citation alignment is evaluated on 20 cases whose gold evidence locus contains an explicit page number, comprising 10 textual and 10 mathematical cases. The remaining 10 visual cases identify a figure or table without specifying a gold page.

On the 20 page-labeled cases, cited-page alignment decreases from 0.670 before repair to 0.618 afterward. The two diagnostics therefore move in different directions: increasing lexical overlap does not necessarily improve spatial alignment with the gold evidence. We consequently treat reference-unit overlap and page-level citation alignment as separate diagnostics rather than optimizing repair against overlap alone.

#### 5.4 Local execution

We measure query-time execution on 100 valid requests using an Apple M3 Pro with 18 GB of unified memory and local Ollama inference. Documents are already prepared for querying, so these measurements cover the answering pipeline rather than PDF ingestion.

Median end-to-end latency is 27.34 s, with  $p_{90} = 41.61$  s and  $p_{95} = 149.54$  s. Retrieval takes a median of 523 ms, although its  $p_{95}$  reaches 112.85 s. Generation takes a median of 26.59 s with  $p_{95} = 37.48$  s, while verification and trace writing together take a median of 15 ms with  $p_{95} = 24$  ms.

The long retrieval tail should not be interpreted as steady-state retrieval cost. Some requests in-

clude first-use embedding or index construction, and cache state was not controlled during these measurements. Stage-wise percentiles are computed independently and therefore do not sum to the corresponding end-to-end percentile.

Overall, the evaluation shows that SCHOLAR can retrieve page-level evidence and execute the complete question-answering and inspection workflow on the tested consumer workstation. Human ratings are generally favorable on average, although agreement between evaluators is limited.

## 6 Availability and Reproducibility

A self-contained [review package](#) provides the application source, setup instructions, dependencies, and an example PDF. A workflow screencast will be available [here](#). Both resources are intended to be accessible without authentication.

## 7 Conclusion

SCHOLAR preserves source identity from PDF ingestion to interface navigation, turning generated citations into inspectable links to source regions. Results show strong page-level retrieval and favorable mean human ratings, while low inter-rater agreement and repair trade-offs bound those findings. The demonstration provides a local workflow for moving directly from generated claims to the evidence used to support them.

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## A Implementation Details

**Parsing and evidence representation.** Docling (Auer et al., 2024) is used when its local assets are available; otherwise ingestion falls back to PyMuPDF. Parser-produced blocks are used directly as retrieval units rather than being divided by a second fixed-length chunker. Each evidence object retains its document identifier, page, modality, textual or caption content, and normalized bounding box. Figures and tables retain associated visual regions, while full-page images are stored separately.

**Retrieval.** Lexical retrieval uses BM25 (Robertson and Zaragoza, 2009), while dense retrieval uses all-MiniLM-L6-v2, based on Sentence-BERT representations (Reimers and Gurevych, 2019). Figure and table intent is detected with keyword, regular-expression, section, and modality heuristics. Ranked channels are combined using

reciprocal-rank fusion (Cormack et al., 2009) with  $k = 60$ . A local cross-encoder reranks the fused set when available, with a deterministic heuristic fallback otherwise. At most seven evidence items are supplied to the generator.

**Generation and citation binding.** Selected evidence is serialized into a structured prompt and assigned temporary identifiers such as [E1]. The application maintains a deterministic mapping from these identifiers to the underlying evidence objects. Generation runs through Ollama (Ollama, 2026), with qwen3.5:9b (Qwen Team, 2026) as the default local model. After generation, returned identifiers are resolved through the stored mapping; unknown identifiers are discarded rather than treated as valid citations.

**Verification and traces.** The verifier combines lexical-overlap variants, polarity checks, and numerical matching while ignoring structural numbers such as figure and table labels. Depending on repair mode, it may remap a citation, narrow a partially supported claim, or remove unsupported material. These checks are not semantic-entailment judgments.

Each request produces a local JSON trace containing the query, retrieval metadata, prompt evidence, generated and normalized answers, citation records, verifier reports and actions, timings, hardware metadata, and final status. The assembled prompt itself is represented by a hash.

**Local execution.** Strict-local mode disables external acquisition features at the application level, restricts model endpoints to local services, and enables offline settings for supported model libraries. This mechanism is application-level rather than an operating-system firewall. If local Docling assets are unavailable, parsing falls back to PyMuPDF rather than downloading them.

## B Evaluation Details

**Human evaluation.** Five evaluators rated the same 45 cases on five-point scales for correctness, completeness, grounding, citation quality, and utility. Inter-rater agreement is reported using ordinal Krippendorff’s  $\alpha$ .

Across 675 citation judgments, 472 (69.9%) are supported, 119 (17.6%) partially supported, 82 (12.1%) unsupported, and 2 (0.3%) unverifiable.

**Automated audit.** The post-hoc audit contains 150 cases and records correctness, completeness, grounding, and citation quality. Correctness is evaluated on 113 reference-eligible cases: 59 correct, 33 partial, 20 incorrect, and one not judgeable.

Across all 150 cases, completeness has 83 positive, 24 partial, 41 negative, and 2 N/A labels. Grounding has 112 positive, 26 partial, 10 negative, and 2 N/A labels. Citation quality has 110 positive, 28 partial, 10 negative, and 2 N/A labels. The retained artifacts do not independently verify the executing judge, so these outputs are treated as diagnostic rather than human ground truth.

**Repair analysis.** Reference-unit overlap F1 is computed on 30 paired answerable cases and increases from 0.159 to 0.224 after repair. Page-level citation alignment is computed on the 20 cases whose gold evidence locus contains an explicit page number, comprising 10 textual and 10 mathematical cases; it decreases from 0.670 to 0.618. The remaining 10 visual cases identify a figure or table without a gold page and are excluded from this page-alignment denominator.

**Latency.** For 100 valid query-time requests on an Apple M3 Pro with 18 GB unified memory, end-to-end latency has median 27.34 s,  $p_{90} = 41.61$  s, and  $p_{95} = 149.54$  s. Retrieval has median 523 ms and  $p_{95} = 112.85$  s; generation has median 26.59 s and  $p_{95} = 37.48$  s; verification plus trace writing has median 15 ms and  $p_{95} = 24$  ms.

Some requests include first-use embedding or index construction, and cache state was uncontrolled. Stage-wise percentiles are computed independently and therefore do not sum to the corresponding end-to-end percentile.